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4. Live Demo

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1. How To Apply

2. Onboarding & Access

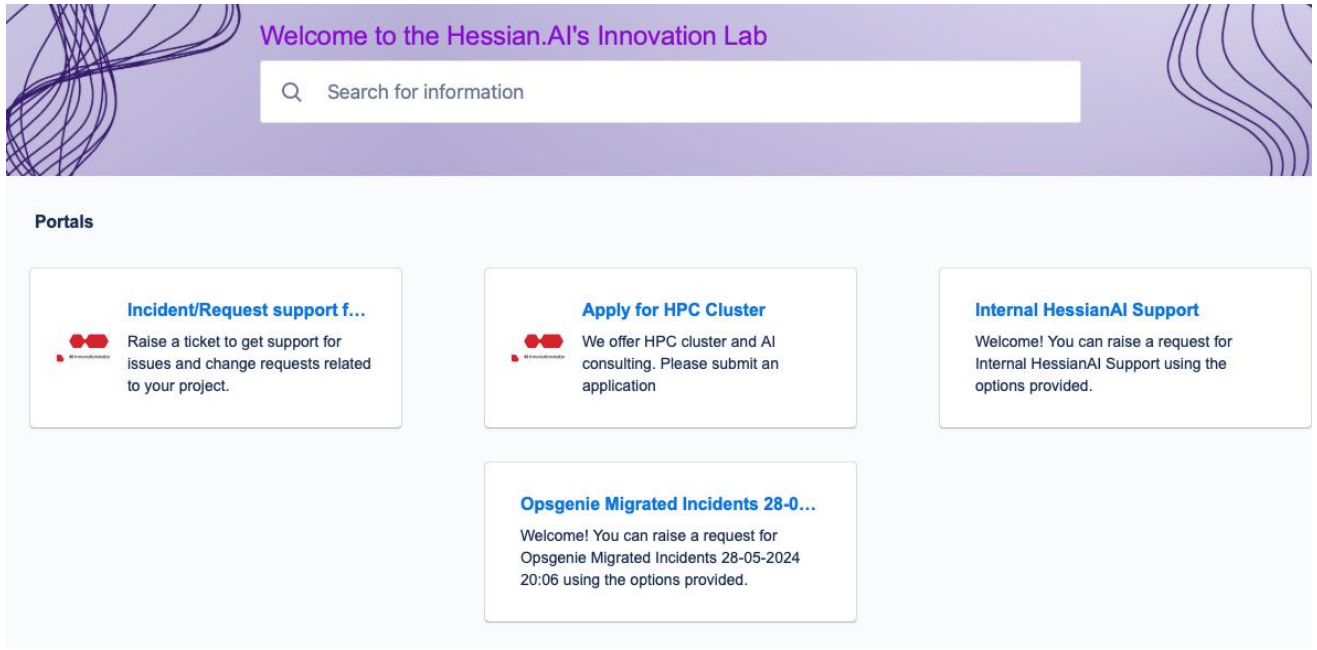
3. Troubleshooting & Support

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Application Form

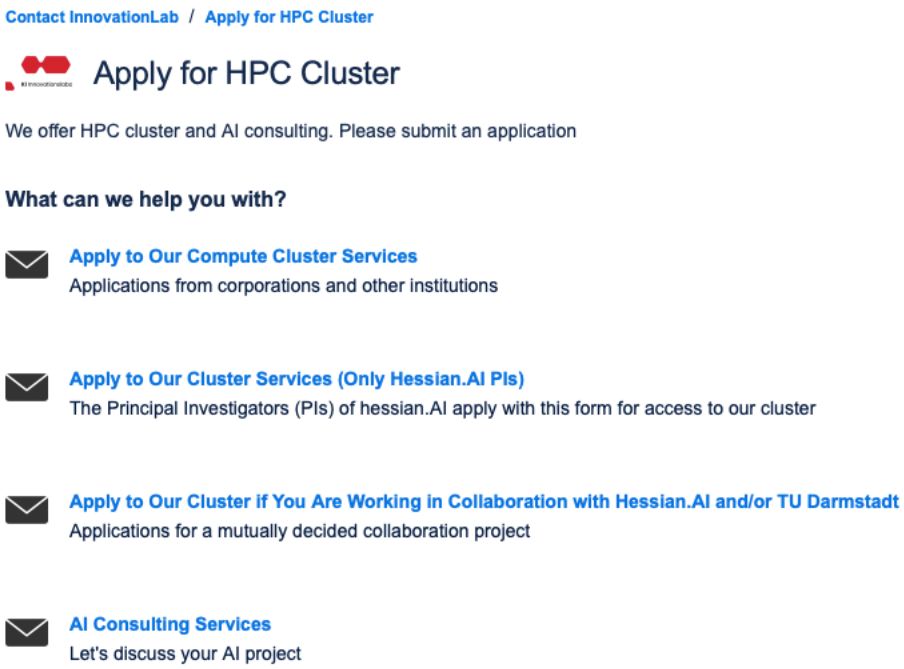
https://hessian-ai.atlassian.net/servicedesk/customer/portals

1. Go to “Apply for HPC Cluster”



Application Form

2. Choose the type of application most relevant to you



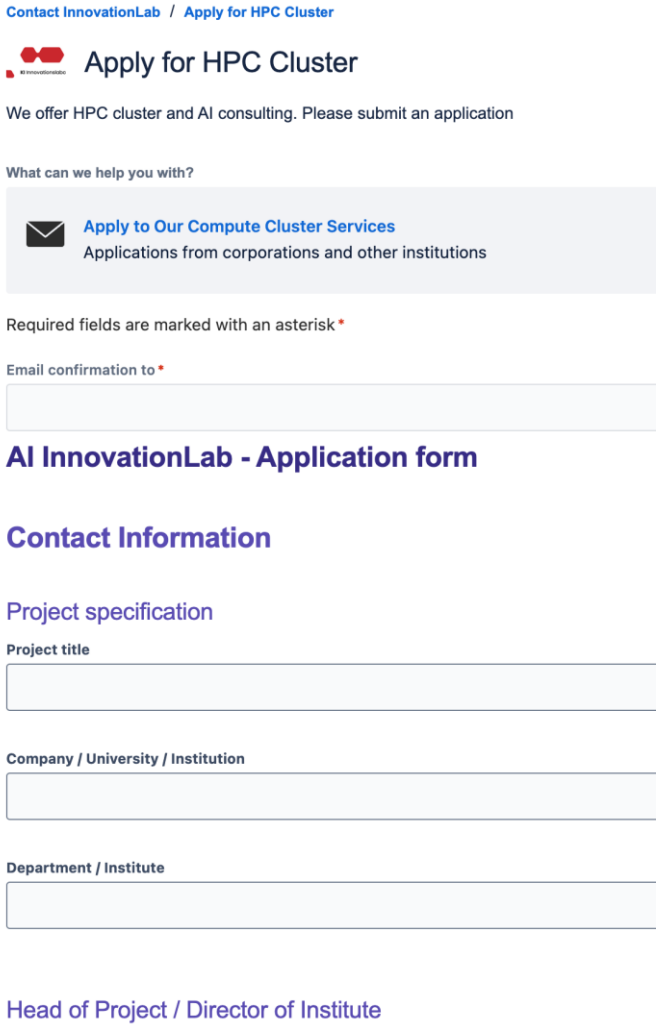
Example Application Form

Please fill out the form as detailed as possible.

This helps reduce overhead time!

Be sure to share your:

- Personal information incl. Email and Phone number
- TU-ID (if you have)
- Project description
- Share your GPU & storage requirements
- Any other specific technical requirements



COMPUTE –Cluster Overview



79 AI compute nodes with 8x NVIDIA A100-80GB GPUs and 1TB RAM each

(SuperPOD, 632 GPUs in total)

Infiniband HDR 800 node interconnect

1,2PB (usable) all flash storage, IBM Spectrum Scale

New Technology Platforms:

- 4 Graphcore M2000 IPUs
- 2 NVIDIA Grace Hopper Dev Kits

COMPUTE – Quotas



If you need extra storage or GPUs (for a short time or permenantly), please contact us by submitting a request here:

https://hessian-ai.atlassian.net/servicedesk/customer/portal/3/group/14/create/38



COMPUTE – Quotas



Each Workspace is restricted to a certain quota, depending on the availability, customer and regulations:

1. GPU Slots: Each workspace has a certain amount of GPUs allocated to it.
- It is up to the users to distribute the slots among their tasks / experiments.

Please remember to SHUT DOWN your experiment once done, so that the GPU is freed and you would not be charged further!

2. Storage Space: Each workspace is allocated 5TB of persistent memory on our cluster.

After Submitting the Application

We will invite you to a first introduction meeting.

In the meantime – if you have any questions, feel free to send a comment in your application ticket.

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Onboarding

1. TU-ID creation

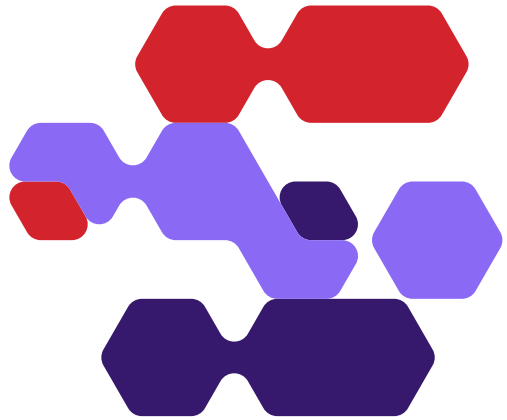
- Follow the steps in the Email from the TU-Darmstadt IT.
- You will then receive a TU-ID and can generate a password.

  [http://idm.tu-darmstadt.de]

- With that TU-ID we create an account in our cluster for you.







Access

TUDA-network / 42-Cluster

- **You need to be in the TU Darmstadt network in order to access the cluster**
- **If you are in the TU Darmstadt use the eduroam network**
- **From outside first connect to the TUDA network using a VPN client e.g. with the Cisco Secure Client**
    - For details regarding the TUDa-VPN, please refer to: https://www.hrz.tu-darmstadt.de/support_und_anleitungen/hrz_anleitungen/tu_vpn/index.de.jsp

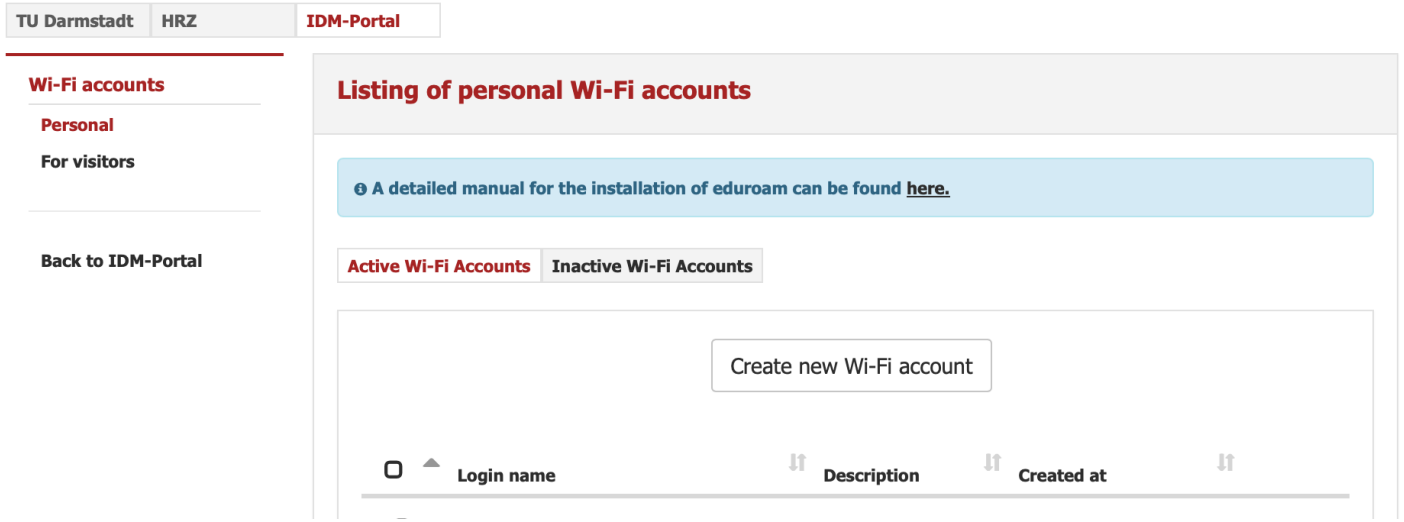
Onboarding

1. TU-ID creation

2. Install VPN & Create TU-Darmstadt connection

   - Follow the tutorial here: https://www.hrz.tu-darmstadt.de/services/it_services/wlan/index.de.jsp

   - Create the VPN connection here (using your TU-ID): https://www.idm.tu-darmstadt.de/wlan/wlan



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Access – MLDE – WebUI

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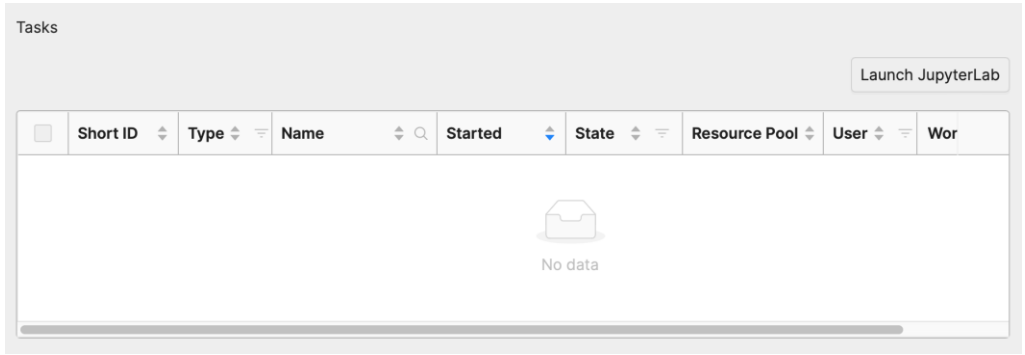
Access – MLDE – WebUI

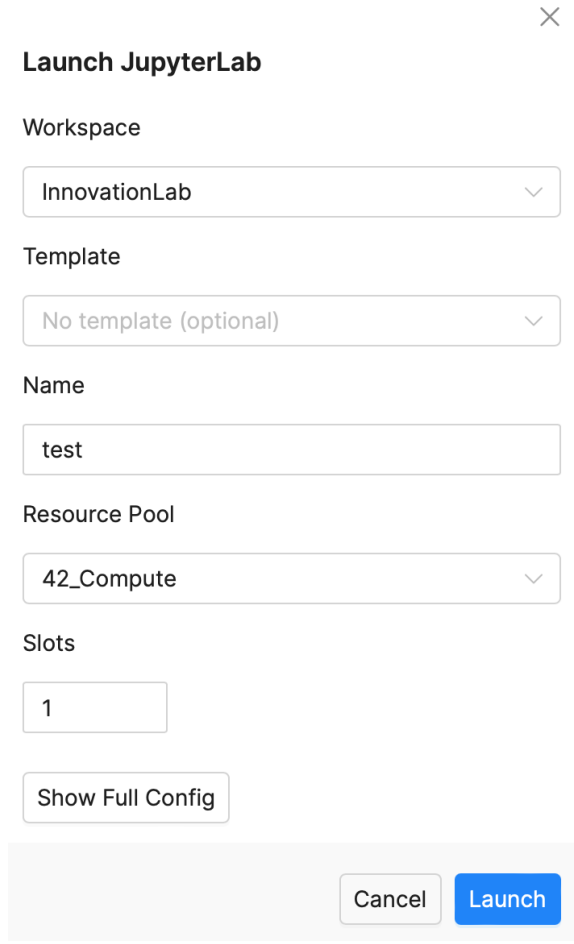
MLDE Web GUI at
https://login01.ai.tu-darmstadt.de:8080

WebUI
(Web User-Interface)

Jupiter Notebook

1) Log into WebUI and select your Worksapce
2) Switch to Task-Tab
3) Click on “Launch JupyterLab”





Access – MLDE – WebUI

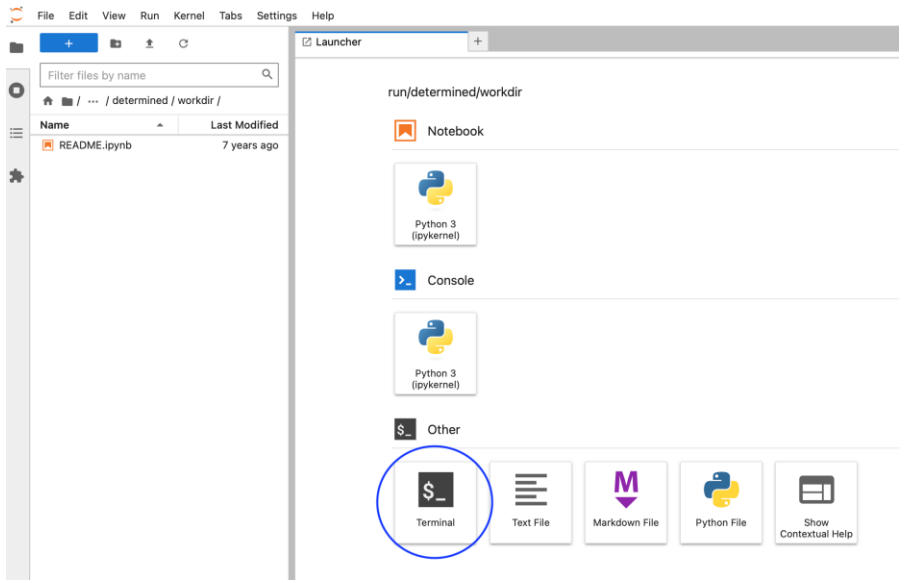
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MLDE Web GUI at
https://login01.ai.tu-darmstadt.de:8080

Shell Container

1) Launch a JupiterLab (see above)
2) Klick on “Terminal”

WebUI
(Web User-Interface)



Access – MLDE – WebUI

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MLDE Web GUI at
https://login01.ai.tu-darmstadt.de:8080



WebUI
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Jupiter Notebook

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Shell Container

1) Launch a JupiterLab (see above)
2) Klick on “Terminal”

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Access – MLDE – Terminal

Access – MLDE - Terminal

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- **Create and activate a virtual environment**

Assuming you have a python version installed, you might want to create a virtual environment first:
Conda: conda create –n myenv then conda activate myenv
Virtualenv: virtualenv myenv then (on Unix) source myenv/bin/activate (on Windows) .\myenv\Scripts\activate

- **Install the determined package**

pip install determined==0.35.0

- **Set the environment variable for DET_MASTER to point to the MLDE interface and log in**

(on Unix) export DET_MASTER=https://login01.ai.tu-darmstadt.de:8080
(on Windows cmd) set DET_MASTER=https://login01.ai.tu-darmstadt.de:8080
(on Windows PowerShell) $env:DET_MASTER="https://login01.ai.tu-darmstadt.de:8080"
Check it with echo %DET_MASTER%
det user login <your TU-ID>

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Access – MLDE - Terminal

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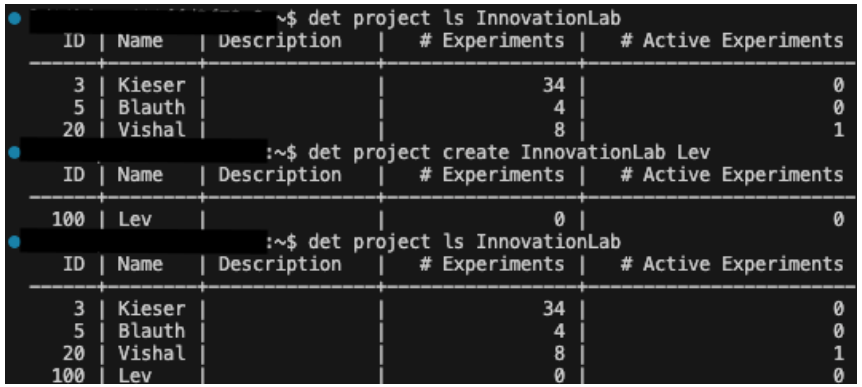
Creating a project

det project ls <workspace_name>

det project create <workspace_name> <project_name>

Note that in the project list you will see the ID of your project, you will need this when you create an experiment

CLI
(Command Line Interface)



Access – MLDE - Terminal

forty 42 two

Install the determined Python package

pip install determined==0.35.0

Set the environment variable for DET_MASTER to point to the MLDE interface:
export DET_MASTER=https://login01.ai.tu-darmstadt.de:8080

det user login <your TU-ID>



CLI
(Command Line Interface)



Jupiter Notebook

det notebook start –w <WorkspaceName>



Shell Container

det shell start -w <WorkspaceName>

Access – MLDE - Terminal

Install the determined Python package

pip install determined==0.35.0

Set the environment variable for DET_MASTER to point to the MLDE interface:
export DET_MASTER=https://login01.ai.tu-darmstadt.de:8080

det user login <your TU-ID>

CLI
(Command Line Interface)

Jupiter Notebook

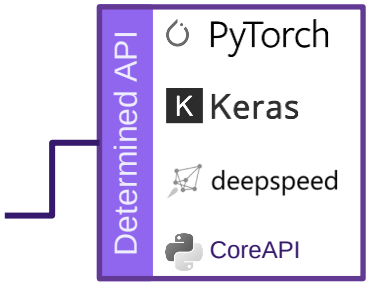
det notebook start –w <WorkspaceName>

Shell Container

det shell start -w <WorkspaceName>

Experiment

det experiment create --project_id <ProjectID> <config_file>



Experiment configuration

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- The behavior of an experiment (also notebooks) is configured via a YAML file.
- You can create your own config file and pass it as an argument to the command.
- Official experiment configuration documentation from determined.AI
- Example config file

One can pass the config file as an argument

```
det notebook start -w <WorkspaceName> --config resources.resource_pool=42_Compute

det notebook start -w <WorkspaceName>  --config-file myconfig.yaml
```

myconfig.yaml:

```
resources:
  resource_pool: 42_Compute
```

Yaml Config: Properties to change

You can download this here
[Example config file]

```
bind_mounts:
  - container_path:
/pfss/mlde/workspaces/mlde_wsp_<Your_workspace_name>
    host_path:
/pfss/mlde/workspaces/mlde_wsp_<Your_workspace_name>
    propagation: rprivate
    read_only: false
  - container_path: /pfss/mlde/users/<Your_TU-ID>
    host_path: /pfss/mlde/users/<Your_TU-ID>
    propagation: rprivate
    read_only: false
debug: false
description: <Your_experiment_description>
entrypoint: null
environment:
  add_capabilities:
    - IPC_LOCK
  drop_capabilities: null
  environment_variables: {}
  force_pull_image: false
  image:
    cpu: determinedai/environments:py-3.8-pytorch-1.12-tf-2.11-
cpu-2b7e2a1
    cuda: determinedai/environments:cuda-11.3-pytorch-1.12-tf-
2.11-gpu-2b7e2a1
    rocm: determinedai/environments:rocm-5.0-pytorch-1.10-tf-
2.7-rocm-2b7e2a1
  pod_spec: null
  ports: null
  proxy_ports: null
idle_timeout: null
notebook_idle_type: kernels_or_terminals
pbs: {}
resources:
  devices:
    - container_path: /dev/infiniband/
      host_path: /dev/infiniband/
      mode: mrw
  resource_pool: 42_Compute
  slots: <Number_of_GPUS>
  weight: 1
slurm: {}
work_dir: /pfss/mlde/workspaces/mlde_wsp_<Your_workspace_name>
```

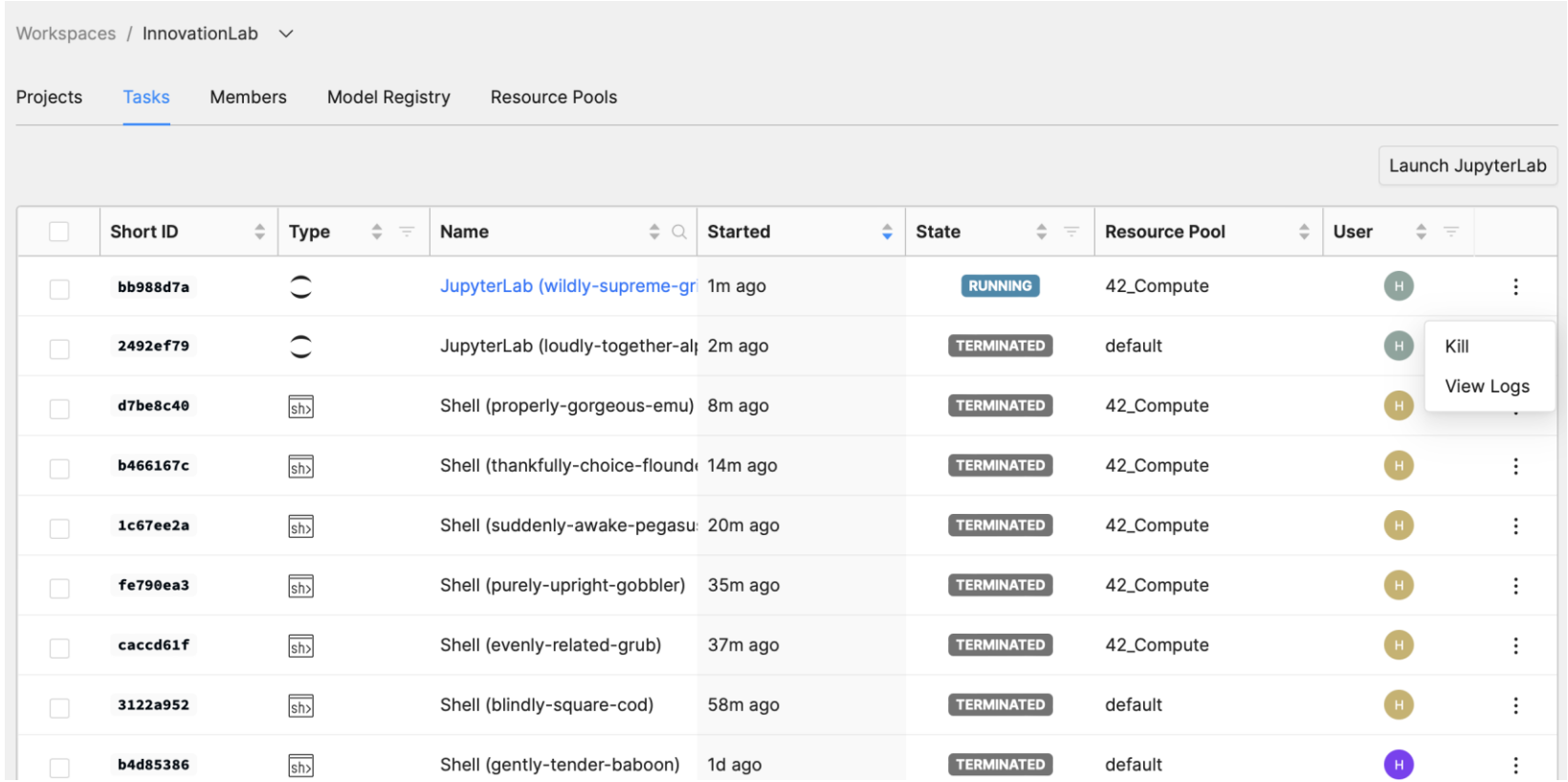
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Stopping a shell

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From the command line:

det shell kill <ID>



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1. Overview of the 42 infrastructure

2. Onboarding & Access

3. Troubleshooting & Support

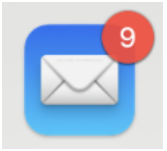
4. Monitoring & Accounting

TroubleShooting & Support

Receive information about cluster

You have been invited to the 42cluster mailing list. Through this mailing list you will get updates on the cluster:

- Downtime
- Version update
- New features



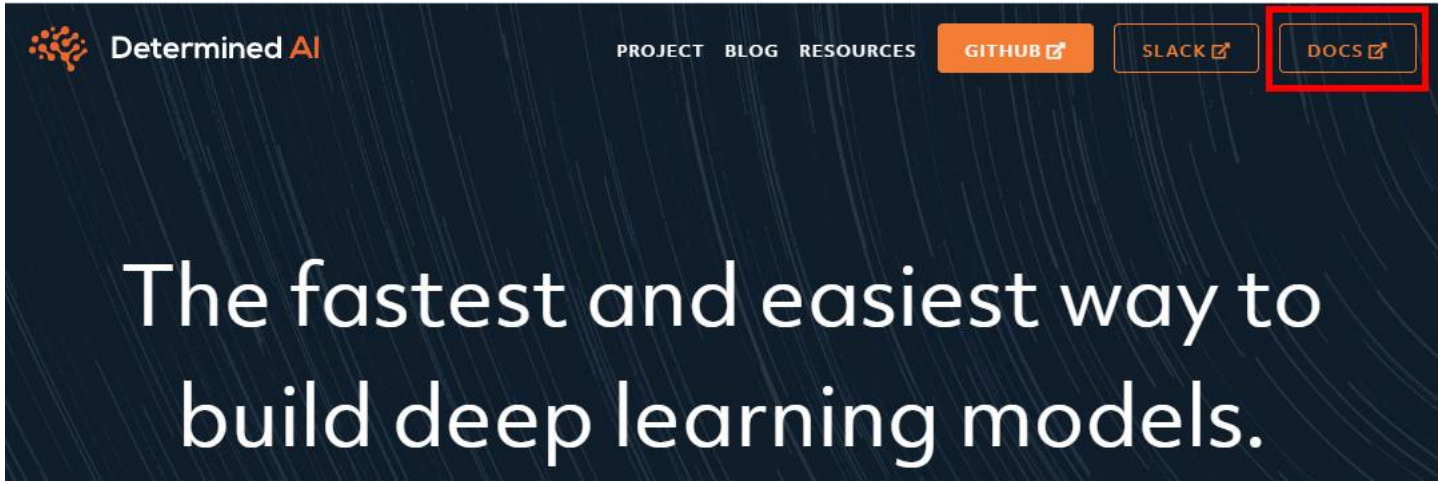
Discuss your ML topics with other teams

Join our Slack workspace to be part of the community: 42-Workspace

TroubleShooting & Support

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Where can I find help & information?



URL: https://docs.determined.ai/0.31.0/index.html

Recommendation: Use one of the examples in the docs as a starting point

TroubleShooting & Support

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What can I do if the documentation does not help?



We do not offer 24/7 support.

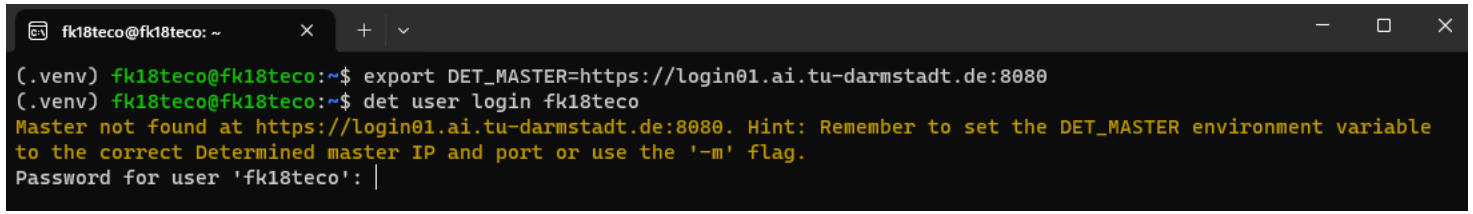
URL: https://hessian-ai.atlassian.net/servicedesk/customer/portal/3

TroubleShooting & Support

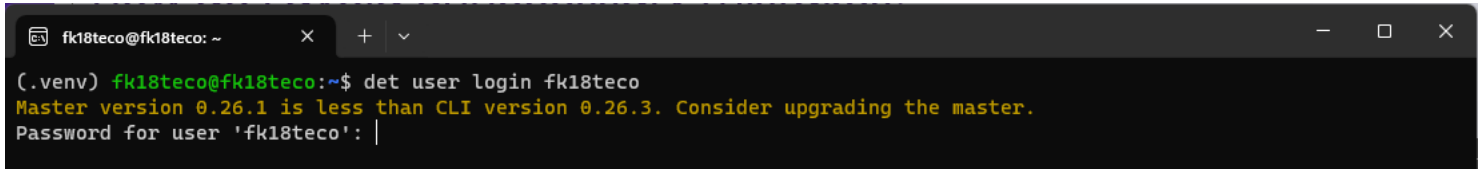
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How do I avoid unnecessary mistakes?

1) Make sure you are connected to the TUDa Network



2) Make sure you have installed the correct Determined version [currently 0.35.0 you see it on the bottom left in the mlde webUI]

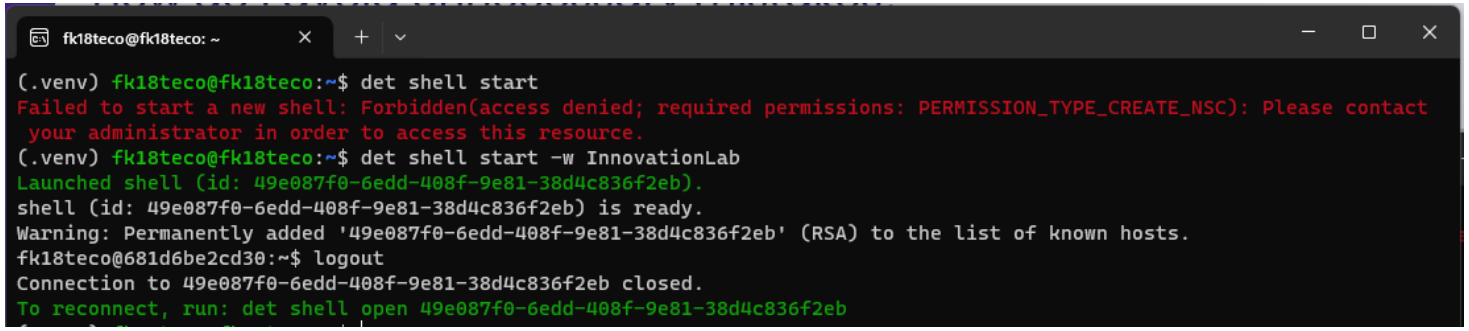


TroubleShooting & Support

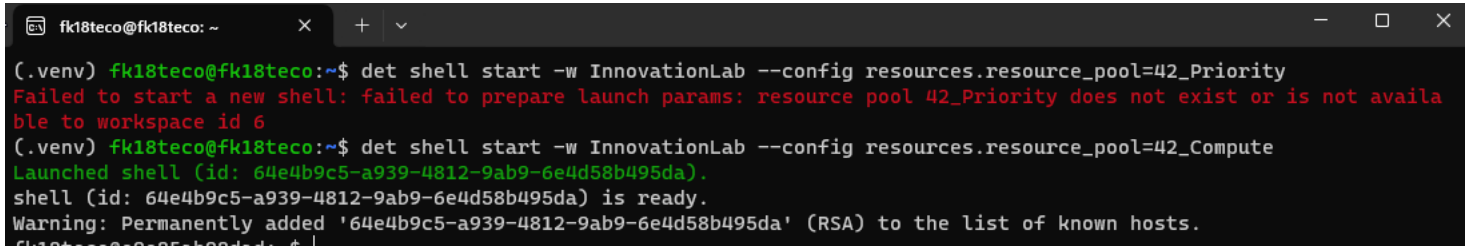
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How do I avoid unnecessary mistakes?

3) Start all experiments / Jupiter labs / shells within your own Workspace



4) Make sure to select the resource pool “42_Compute”



TroubleShooting & Support

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How do I avoid unnecessary mistakes?

5) Stick to the structure and indentations of the experiment configuration

```
1  bind_mounts:
2    - container_path: /pfss/mlde/mlde_ext
3      host_path: /pfss/mlde/mlde_ext
4      propagation: rprivate
5      read_only: false
6    - container_path: /pfss/mlde/users/fk18teco
7     host_path: /pfss/mlde/users/fk18teco
8
9  debug: false
10 description: JupyterLab (implicitly-curious-cheetah)
11 entrypoint: null
12 environment:
13   add_capabilities:
14     - IPC_LOCK
15   drop_capabilities: null
16   environment_variables: {}
17   force_pull_image: false
18   image:
19     cpu: determinedai/environments:py-3.8-pytorch-1.12-tf-2.11-cpu-2b7e2a1
```

Invalid YAML on line 6.

TroubleShooting & Support

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How do I avoid unnecessary mistakes?

6) Define your environment in your own docker container and load it via the experiment configuration file

```
image:
  cpu: determinedai/environments:py-3.8-pytorch-1.12-tf-2.11-cpu-2b7e2a1
  cuda: determinedai/environments:cuda-11.3-pytorch-1.12-tf-2.11-gpu-2b7e2a1
  rocm: determinedai/environments:rocm-5.0-pytorch-1.10-tf-2.7-rocm-2b7e2a1
```

Note: you are not allowed to install anything on the computational nodes, apt install, pdsh, apt-get won't work, even if it's in the documentation)

TroubleShooting & Support

forty4 2 two

How do I avoid unnecessary mistakes?

7) Close all JupiterLabs and Shell once finished, or start it with 0 GPUs

8) Save all your data (checkpoints, augmented training data, results etc.) on the persistent storage (PFSS) and not on the instance.

Any data in an instance will be deleted once you close it!

The instance has a limited storage space, and if you overuse it, your compute node will crash! That means that all your progress and data is gone, and your instance (and potentially other instances on the same node) will be gone!

Thank you!

Any questions?

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Optional: **set remote connection** (e.g. on VS code)

1. start a shell instance:

det shell start -w <workspace name> --config-file <path to>/config.yaml

2. use the shell ID to create an ssh:

copy shell id (you will see a line in green, Launched shell (id: <copy this!>).

3. Open a new terminal, and there type:

det shell show-ssh-command <paste shell id>

4. Add it via VSCode

Copy the ssh code, go to the command pallet (ctrl+shift+p) and type “connect to host (remote-SSH)”, then type “Add New SSH Host” paste the ssh code and tap enter. This should then add an entry to your .ssh config.



PSA: Managing inodes

- **What are inodes**

Inodes are limited and are basically consumed by files and directories as they store metadata related to them. It is possible to increase the number of available inodes but this can‘t be done indefinitely and our cluster already has a high number of available inodes.

If many of the available inodes are allocated this can impact the whole cluster!

We check the allocated inodes regularly, but ask you to please follow some best practices.

- **How to track your inode usage**

1. Go to the directory you want to check, e.g. cd /pfss/mlde/workspaces/mlde_wsp_<your-workspace>/<your-folder>
2. Check with find . | wc –l or for dir in */; do echo -n "$dir: "; find "$dir" -printf '.' | wc -c; echo " inodes"; done (might take a while)

You should aim to not have more than 2 million inodes (ideally as low as possible).

- **How to reduce your inode usage**

1. Delete files or move them to another storage (**pfss is not meant for long-term storage!)**
2. Transform batches of your data into .tar files (and optionally compress them into .tar.gz) and remove the original files

More information can for example be found here: https://docs.rackspace.com/docs/what-are-inodes-in-linux



Live Demo

https://hpe-mlde.determined.ai/latest/get-started/webui-qs.html#create-your-first-experiment

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